$$\frac{dy}{dt} = v_y = y - \text{component of velocity}$$

$$\frac{dz}{dt} = v_z = z - \text{component of velocity}$$

The magnitude of velocity $v$ is called speed and is given by

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

Speed is always a positive scalar. The unit of speed is also meter per second.

## EXAMPLE 2.21

The position vector of a particle is given $\vec{r} = 2t\,\hat{i} + 3\,t^2\,\hat{j} - 5\,\hat{k}$.

a) Calculate the velocity and speed of the particle at any instant $t$
b) Calculate the velocity and speed of the particle at time $t = 2$ s

### Solution

The velocity $\vec{v} = \frac{d\vec{r}}{dt} = 2\hat{i} + 6t\hat{j}$

The speed $v(t) = \sqrt{2^2 + (6t)^2}\ ms^{-1}$

The velocity of the particle at $t = 2$ s

$$\vec{v}(2sec) = 2\hat{i} + 12\hat{j}$$

The speed of the particle at $t = 2$ s

$$v(2\,s) = \sqrt{2^2 + 12^2} = \sqrt{4 + 144}$$
$$= \sqrt{148} \approx 12.16\,m\,s^{-1}$$

Note that the particle has velocity components along x and y direction. Along the z direction the position has constant value (–5) which is independent of time. Hence there is no z-component for the velocity.

## EXAMPLE 2.22

The velocity of three particles A, B, C are given below. Which particle travels at the greatest speed?

$$\overrightarrow{v_A} = 3\hat{i} - 5\hat{j} + 2\hat{k}$$
$$\overrightarrow{v_B} = \hat{i} + 2\hat{j} + 3\hat{k}$$
$$\overrightarrow{v_C} = 5\hat{i} + 3\hat{j} + 4\hat{k}$$

### Solution

We know that speed is the magnitude of the velocity vector. Hence,

$$\text{Speed of A} = \left|\overrightarrow{v_A}\right| = \sqrt{(3)^2 + (-5)^2 + (2)^2}$$
$$= \sqrt{9 + 25 + 4} = \sqrt{38}\ \text{m s}^{-1}$$

$$\text{Speed of B} = \left|\overrightarrow{v_B}\right| = \sqrt{(1)^2 + (2)^2 + (3)^2}$$
$$= \sqrt{1 + 4 + 9} = \sqrt{14}\ \text{m s}^{-1}$$

$$\text{Speed of C} = \left|\overrightarrow{v_C}\right| = \sqrt{(5)^2 + (3)^2 + (4)^2}$$
$$= \sqrt{25 + 9 + 16} = \sqrt{50}\ \text{m s}^{-1}$$

The particle C has the greatest speed.

$$\sqrt{50} > \sqrt{38} > \sqrt{14}$$

## EXAMPLE 2.23

Two cars are travelling with respective velocities $\vec{v}_1 = 10\,m\,s^{-1}$ along east and $\vec{v}_2 = 10\,m\,s^{-1}$ along west. What are the speeds of the cars?

### Solution

Both cars have the same magnitude of velocity. This implies that both cars travel at the same speed even though they have



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